Name: Mitul Marimuthu *(as it would appear on official course roster)*	
UCSB email address: mmarimuthu	**@ucsb.edu**

Homework 01: Delay and Loss, HTTP, DNS, Getting Started with Wireshark
Points: *100*

- MAY ONLY BE TURNED IN ON **GRADESCOPE as a PDF file.**
- There is NO MAKEUP for missed assignments.
- We are strict about enforcing the LATE POLICY for all assignments (see syllabus).
  Only use the space provided for answers. Use clear and clean handwriting (or typing).
  NOT USING THIS TEMPLATE WILL LOSE YOU 20 POINTS!

---

1. **(10 pts) Delays.** Future Mars inhabitants may still need to post TikTok videos, but how well will the network work for them? Assume there are two options for Earth-Mars communication: directly, or via a store-and-forward router on the moon. The individual links are shown. Assume the speed of light is 300,000 km/sec.

a. **(3 pts)** Assume that video compression technology is quite good by this point, and a Mars resident only needs to send 1.5 Mb to Earth to upload their video. If they send it via the direct path, how long will it take to send the entire video?

$D_{total} = D_{prop} + D_{trans}$

$D_{trans} = \frac{1.5 \text{ Mb}}{100 \text{ bps}} = 1.5 \cancel{\text{Mb}} \cdot \frac{1 s}{100 \cancel{b}} \cdot \frac{1000000 \text{ b}}{\cancel{\text{Mb}}} = 15,000 \text{ s}$

$D_{prop} = \frac{94.6 \text{ m km}}{300,000 \text{ km/s}} = 94.6 \times 10^6 \cancel{\text{km}} \cdot \frac{1 s}{300,000 \cancel{\text{km}}} = 315.33 \text{ s}$

$\boxed{D_{total} = 15,315.33 \text{ s}}$

b. **(3 pts)** How long will it take to send the same 1.5 Mb from Mars to the moon?

$$D_{trans} = \frac{1.5\ Mb}{500\ bps} \cdot 1.5\ \cancel{Mb} \cdot \frac{1\ sec}{500\ b} \cdot \frac{1{,}000{,}000\ b}{\cancel{Mb}} = 3000\ s$$

$$D_{prop} = \frac{94.2 \times 10^6\ km}{3 \times 10^5\ km/s} = 94.2 \times 10^6\ km \cdot \frac{1\ sec}{3 \times 10^5\ s} = 314\ s$$

$$\boxed{D_{tot} = 3314\ s}$$

c. **(4 pts)** How long will it take to send the 1.5 Mb to Earth from Mars via the moon?

$$D_{trans} = \frac{1.5\ Mb}{9\ Mbps} = 1.5\ \cancel{Mb} \cdot \frac{1\ s}{9\ \cancel{Mb}} = 0.167\ s$$

$$D_{prop} = \frac{384{,}000\ km}{300{,}000\ km/s} = 1.28\ s$$

$$D_{tot_2} = 1.45\ s$$

$$D_{tot} = 3314\ s + 1.45\ s = \boxed{3315.45\ s}$$

2. **(24 pts) Delays.** Two hundred packets, each 5 Kb in size, are sent from host A to host B across a network consisting of 5 store-and-forward switches. The network topology is:

A — S1 — S2 — S3 — S4 — S5 — B

All 6 links are identical: 5 Mbps bandwidth and 200 m long. Assume no acknowledgments, the propagation speed is 300 meters per microsecond, and nodal processing delay is zero.

a. **(3 pts)** What is the total transmission delay required to push all 200 messages onto the network?

$$D_{trans_1} = \frac{5\,kb}{5\,Mbps} = \cancel{5}kb \cdot \frac{1\,s}{\cancel{5}\,Mb} \cdot \frac{1\,Mb}{1000\,kb} = 0.001\,s$$

$$D_{TOTAL} = 200 \cdot D_{trans_1} = \boxed{0.2\,s}$$

b. **(3 pts)** What is the total end-to-end propagation delay across the entire physical path?

$$D_{prop} = 6 \cdot \frac{200\,m}{300\,m/\mu s} = 6 \cdot 200\,m \cdot \frac{1\,\mu s}{300\,m} = \frac{12}{3} = \boxed{4\,\mu s}$$

c. **(2 pts)** What is the total queuing delay?

$D_{queue} = 0$ because all the links have the same bandwidth so there are no bottlenecks in the network and because there is only one host sending data

d. **(3 pts)** What is the total store and forward delay?

$$D_{snf} = \sum_{i=1}^{5} D_{trans_1} = \sum_{i=1}^{5} 0.001 = \boxed{0.005\,s} = 5\,ms$$

e. **(4 pts)** What is the total delay from the time host A begins transmitting the first packet until host B finishes receiving the last packet? Account for pipelining across the switches.

$D_{trans} = 0.2\ s \rightarrow$ time it takes until the last packet gets into the network

$D_{across} = D_{snf} + D_{prop} = 0.005 + 4\ \mu s = 0.005004\ s$

$\boxed{D_{total} = 0.205004\ s}$

f. **(2 pts)** What's the maximum achievable end-to-end throughput from host A to host B?

5 Mbps $\Rightarrow$ no bottlenecks

g. **(3 pts)** What is the average end-to-end throughput for this transfer?

$T_{avg} = \frac{200.5\ kb}{0.205\ secons} = \boxed{4.88\ Mbps}$

h. **(4 pts)** Suppose host C is connected to switch S1 by an identical link and also sends 200 packets of 5 Kb simultaneously with host A, all destined for host B. Which of the delays from parts (a) - (e) would change? Calculate the new total end-to-end delay.

Queueing delay increases significantly
Total end-to-end delay also increases
400 total packets being sent now

$D_{trans_{S1}} = 400 \cdot \frac{5\ kb}{5\ Mbps} \cdot \frac{1\ Mb}{1000\ kb} = 0.4\ s$

$D_{across} = 0.001\ s/\text{packet}/\text{switch} \cdot 5\ \text{switches} + 4\ \mu s$ (prop delay) $= 0.005004\ s$

$\boxed{D_{total} = 0.405004\ s}$

3. **(4 pts) HTTP.** A webpage contains 10 images. How many RTTs does it take a browser to download and display this webpage, including all embedded/referenced objects: (Do not count the TCP connection close messages.)

a. **(2 pts)** using persistent HTTP without pipelining and a single TCP socket?

persistent -> connection stays open

2 RTTs for the webpage (HTML file)

10 * 1 RTT for each subsequent

$\boxed{12 \text{ total RTTs}}$

b. **(2 pts)** using persistent HTTP with pipelining and a single TCP socket?

sends all 10 image requests at once

1 RTT to establish the connection (handshake

1 RTT to get the webpage

1 RTT to fetch the images

3 total RTTs

4. **(6 pts) HTTP.** An HTTP client needs five objects (two large and three small) to display a webpage. It knows the URL of all objects. How many RTTs will it take for the client to receive the five objects using:

↳ no need for initial fetch of HTML file

a. **(2 pts)** HTTP/1.0?

2 RTTs needed for each object

-> 10 RTTs

b. **(2 pts)** HTTP/1.1 without pipelining?

1 RTTs for handshake

5 * 1 RTT for subsequent 4 objects

-> 6 RTTs

c. **(2 pts)** HTTP/1.1 with pipelining?

Sends all files in the same trip

2 total RTTs

1 RTT for handshake

1 RTT for get request

5. **(9 pts) HTTP.** A web page contains 4 small images, 2 large images, and 1 JavaScript file.

a. **(3 pts)** How many messages and RTTs does it take a browser to download and display this Webpage, including all embedded/referenced objects, using persistent HTTP with pipelining and a single TCP socket?
Note: Do not count the TCP connection close messages. A round trip to the server includes two messages (the request message and the response message).

opening TCP connection => 1 RTT, 2 messages

getting the webpage => 1 RTT, 2 messages

getting all the objects => 1 RTT, 14 messages – 7 requests – 7 responses

Total: 2 RTTs, 18 messages

b. **(3 pts)** The browser can open multiple TCP sockets and use pipelined HTTP requests. How many sockets should it open to minimize the number of RTTs?

Since we're pipelining, opening more sockets won't change the number of RTTs needed. It may be more efficient to account for HOL blocking from the large images, but that has no effect on the RTTs.

$\boxed{\text{1 socket.}}$

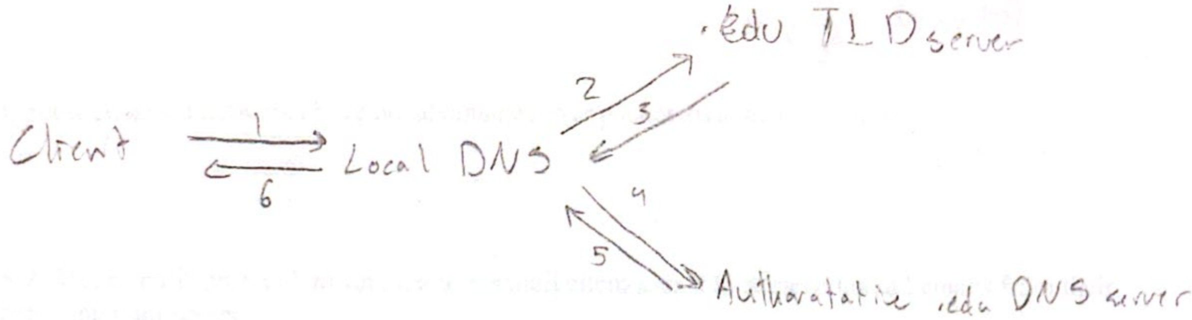
c. **(3 pts)** The browser can open multiple TCP sockets and use pipelined HTTP requests. If the bottleneck link is the client's first link, and this is the only active client on the network, how many sockets should the browser open to minimize the total time to load this webpage?

The client's first link is the link to their Wi-Fi. Opening more sockets won't solve the issue of a slow connection, so the optimal amount of sockets to open is still 1.

6. (16 pts) DNS.

a. **(4 pts) A client needs to resolve the IP address for a completely new .edu web server. Assuming the client's local DNS resolver has the .edu TLD servers cached from normal use (but nothing else regarding this specific domain), how many total messages and RTTs are required to resolve the address?**

3 RTTs, 6 messages



b. **(4 pts) If a client's primary local DNS server crashes and becomes unavailable, what is the most likely effect on the client? Are there any standard network configurations that might allow the client to seamlessly continue resolving names despite the crash?**

If the local DNS server crashes, then the client cannot resolve any domain names. The only way the client can seamlessly continue resolving names is if there is a backup local DNS server or if they already know the IP addresses of the names they want to resolve.

c. **(4 pts)** An organization's authoritative DNS server is located behind a very low-bandwidth network link. Should the organization set high or low TTL (Time-To-Live) values for its DNS records? Briefly explain your reasoning.

They should set high TTL because then records can be cached for a longer time and queries don't have to frequently be sent across the low bandwidth link to the authoritative server. This improves latency & prevents congestion.

d. **(4 pts)** A host makes two sequential DNS requests for the same 'A' record. Will it always receive the exact same IP address in response? Provide at least two network or server-side reasons why the answer might differ.

No, it will not always receive the same IP address

1. If the TTL is 0 and the IP address changes in between requests, the returned IP will be different

2. Round Robin DNS ⇒ to make sure a server doesn't get overwhelmed, a single hostname can point to multiple different servers and the authoritative DNS server can perform round-robin to balance the load among them, resulting in different IP addresses after subsequent queries.

7. **(16 pts) True or False.** Indicate whether each statement is **True (T)** or **False (F)**.

a. DNS can resolve the same domain name to multiple IP addresses, and DNS can resolve different domain names to the same IP address.

T

b. Routers use DNS MX records and the SMTP protocol to route an email to the receiver's mail server.

F

c. Circuit-switched networks have no advantages over packet-switched networks.

F

d. SMTP is a "pull" protocol, meaning a user's mail client uses it to retrieve unread emails from their receiving mail server.

F

e. BitTorrent incentivizes users to upload data through a "tit-for-tat" mechanism, where a peer will unchoke (send data to) peers that are currently sending it data at the highest rates.

T

f. The Internet's DNS root servers store the IP addresses of every individual website (like www.google.com or www.wikipedia.org) directly in their databases.

F

g. Packet loss typically occurs when a packet arrives at a router, but the router's buffer (queue) is completely full, causing the router to drop the newly arrived packet.

T

h. HTTP is considered a stateless protocol because the server itself does not retain any memory of previous requests made by the same client.

T

(15 pts) Wireshark Lab Handin
Please follow the Wireshark lab writeup and provide your answers to the following questions.

1. **(5 pts)** List the different protocols that appear in the protocol column in the unfiltered packet-listing window in step 3 above.

ICMP, ARP, DNS

2. **(5 pts)** How long did it take from when the HTTP GET message was sent until the HTTP OK reply was received? (By default, the value of the Time column in the packet-listing window is the amount of time, in seconds, since Wireshark tracing began. To display the Time field in time-of-day format, select the Wireshark *View* pull down menu, then select Time *Display Format*, then select *Time-of-day*.)

21:40:55.217123279
− 21:40:55.148532004

$\boxed{0.068591275 \text{ s}}$

3. **(5 pts)** What is the Internet address of the gaia.cs.umass.edu (also known as www-net.cs.umass.edu)?

→ IP address

128.119.245.12